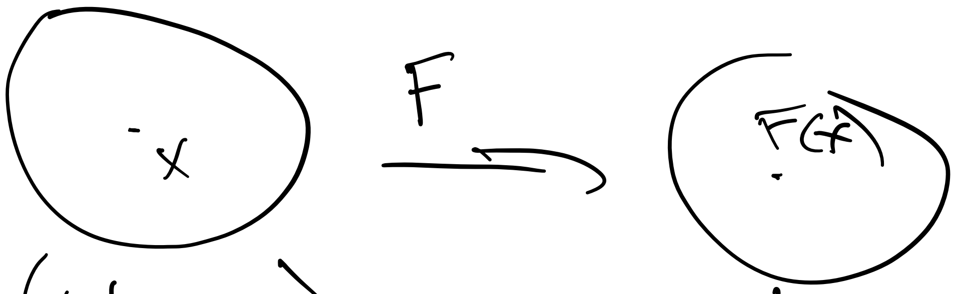
Семинар 11.

Матрично-векторное дифф-е.



$(U, \|\cdot\|_U)$ конечномерно $\qquad$ $(V, \|\cdot\|_V)$ конечномерно

Опр. Пусть $f: U \to V$, $x \in U$, тогда ф-я наз-ся дифф в $(\cdot)\, x$ с дифф-лом $df(x)$, если

$$\underset{\in V}{F(x+h)} = \underset{\in U}{F(x)} + \underbrace{df(x)[h]}_{\in V} + o(\|h\|), \quad \|h\| \to 0$$

1) $\mathbb{R} \to \mathbb{R}$

$$F(x+h) - F(x) = A_x \cdot h + o(|h|)$$

2) $\mathbb{R}^n \to \mathbb{R}$

$$F(x+h) - F(x) = \langle \nabla F(x), h \rangle + o(\|h\|)$$

$$\nabla F(x) = \left(\frac{\partial F}{\partial x_1}(x) \quad \dots \quad \frac{\partial F}{\partial x_n}(x)\right)^T$$

3) $\mathbb{R}^{n\times n} \to \mathbb{R}$

$$F(x+h) - F(x) = \underbrace{\langle \nabla F(x), H \rangle}_{tr(\nabla F(x)^T \cdot H)} + o(\|H\|)$$

$$\nabla F(x) = \begin{pmatrix} \frac{\partial F}{\partial x_{11}}(x) & \dots & \frac{\partial F}{\partial x_{1n}}(x) \\ \vdots & & \\ \frac{\partial F}{\partial x_{n1}}(x) & \dots & \frac{\partial F}{\partial x_{nn}}(x) \end{pmatrix}$$

$$\|X\|_F = \sqrt{\langle X, X \rangle_F} = \sqrt{tr(X^T X)} =$$

$$= \sqrt{\begin{pmatrix} x_{11} & \dots & x_{1n} \\ \vdots & & \\ x_{n1}, & \dots & x_{nn} \end{pmatrix}^T \cdot \begin{pmatrix} \dots \end{pmatrix}} =$$

$$\sqrt{tr\left(diag\left(x_{11}\cdot x_{11} + x_{12}\cdot x_{21} \dots \quad , \dots \right)\right)}$$

4) $F: \mathbb{R}^n \to \mathbb{R}^m$

$$F(x+h) - F(x) = J_F \cdot h + \bar{o}(\|h\|$$

$$y_F = \begin{pmatrix} \nabla F_1(x)^T \\ \vdots \\ \nabla F_n(x)^T \end{pmatrix} = \begin{pmatrix} \frac{\partial F_1}{\partial x_1}(x), \dots \frac{\partial F_1}{\partial x_n}(x) \\ \dots \\ \dots \end{pmatrix}$$

$$dF(x)[h] = \langle \nabla F(x), h \rangle = \underbrace{\boxed{\nabla F(x)^T}}_{y_F} h$$

g \ f	$\mathbb{R}$	$\mathbb{R}^m$
$\mathbb{R}$	$F'(x) \cdot h$	—
$\mathbb{R}^n$	$\langle \nabla, h \rangle$	$y_F \cdot h$
$\mathbb{R}^{m \times n}$	$\langle \nabla, H \rangle$	—

1) $d(\alpha X) = \alpha dX$
2) $d(AXB) = A\,dX\,B$
3) $d(F(x) + G(x)) = dF(x) + dG(x)$
4) $dX^T = (dX)^T$
5) $d(\langle F(x), G(x) \rangle) = \langle dF(x), G(x) \rangle + \langle F(x), dG(x) \rangle$
6) $d\frac{F(x)}{G(x)} = \frac{dF(x)\,G(x) + F(x)\,dG(x)}{G(x)^2}$

7) $d(\det X) = \mathrm{tr}(\det X \, X^{-1} \cdot dX)$

8) $dX^{-1} = -X^{-1} dX X^{-1}$

9) $dF(X) = F(dX)$

$$d(F \circ G(x))[H] = dF(G(x))[dG(x)[H]]$$

Задача

1) $F(x) = \langle Ax, x\rangle + \langle b, x\rangle + c$

$F: \mathbb{R}^n \to \mathbb{R}$, $\nabla - ?$

$$dF(x) = d(\langle Ax, x\rangle + \langle b, x\rangle + c) =$$

$$= d\langle Ax, x\rangle + d b^T x + \cancel{dc}^{\,0} =$$

$$= \langle A dx, x\rangle + \langle Ax, dx\rangle + b^T dx =$$

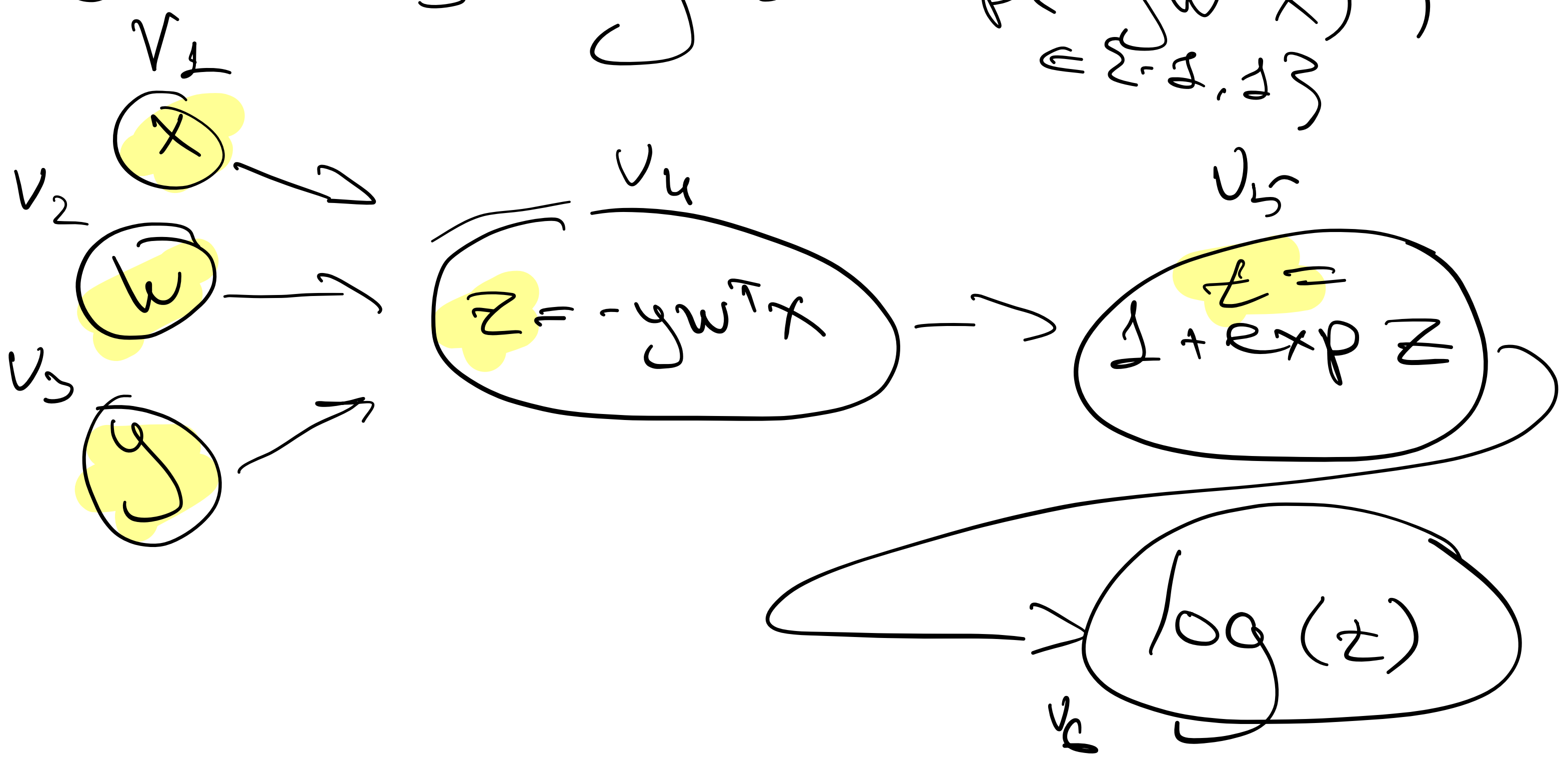
$$= x^T A dx + x^T A^T dx + b^T dx =$$

$$= \langle \underbrace{(A^T + A)x + b}_{\nabla F(x)}, dx\rangle$$

Back propagation

$$f(x, w, y) = \log\left(1 + \exp(-y w^T x)\right)$$

$y \in \{-1, 1\}$



forward pass

backward pass

forward accum

backward accum

Backward accumulation

1) Нумерация вершин
2) $\dfrac{\partial f}{\partial V_{Last}} = 1$

3) $$\frac{\partial f}{\partial v_i} = \sum_{j \in \text{родители}(i)} \frac{\partial f}{\partial v_j} \cdot \frac{\partial v_j}{\partial v_i}$$

$$J^f_{v_i} = \sum_j J^f_{v_j} \, J^{v_j}_{v_i}$$

Задача

1) $$\frac{\partial f}{\partial v_6} = 1$$

2) $$\frac{\partial f}{\partial v_5} = \frac{\partial f}{\partial v_6} \cdot \frac{\partial v_6}{\partial v_5} = \frac{1}{t} = \frac{1}{1+\exp(z)}$$

3) $$\frac{\partial f}{\partial v_4} = \frac{\partial f}{\partial v_5} \cdot \frac{\partial v_5}{\partial v_4} = \frac{1}{1+e^z} \cdot e^z$$

4) $$\frac{\partial f}{\partial v_3} = \frac{\partial f}{\partial v_4} \cdot \frac{\partial v_4}{\partial v_3} = \frac{e^z}{1+e^z} \cdot (-w^T x)$$

5) $$\frac{\partial f}{\partial v_2} = \frac{\partial f}{\partial v_4} \cdot \frac{\partial v_4}{\partial v_2} = \frac{e^{-yw^Tx}}{1+e^{-yw^Tx}} \cdot (-yx)$$

$\frac{\partial f}{\partial v_1}$, $\frac{\partial f}{v_3}$ аналогично

Forward accumulation

1) Пронумеровать вершины

2) $$\frac{\partial v_1}{\partial v_1} = 1$$

3) $$\frac{\partial v_i}{\partial v_1} = \sum_{j \in \text{дети}(i)} \frac{\partial v_i}{\partial v_j} \cdot \frac{\partial v_j}{\partial v_1}$$

Задача

1) $$\frac{\partial v_2}{\partial v_2} = 1$$

2) $$\frac{\partial v_4}{\partial v_2} = \underset{0}{\frac{\partial v_4}{\partial v_1}} \cdot \overset{0}{\frac{\partial v_1}{\partial v_2}} + \frac{\partial v_4}{\partial v_2} \cdot \frac{\partial v_2}{\partial v_2} + \underset{0}{\frac{\partial v_4}{\partial v_3}} \cdot \frac{\partial v_3}{\partial v_2} = -yx$$

В чём подвох?

$$F(x) = D(C(B(A(x)))) : \mathbb{R}^n \to \mathbb{R}^m$$

$$x \to \underbrace{A}_{a} \to \underbrace{B}_{b} \to \underbrace{C}_{c} \to \underbrace{D}_{y}$$

morga $\frac{\partial F}{\partial x} = \frac{\partial y}{\partial c} \cdot \frac{\partial c}{\partial b} \cdot \frac{\partial b}{\partial a} \cdot \frac{\partial a}{\partial x}$

Backward accum

$$\frac{\partial F}{\partial x} = \left(\left(\left(\frac{\partial y}{\partial c}\right)_1 \cdot \frac{\partial c}{\partial b}\right)_2 \cdot \frac{\partial b}{\partial a}\right)_3 \cdot \frac{\partial a}{\partial x}$$

Forward accum

$$\frac{\partial F}{\partial x} = \frac{\partial y}{\partial c} \cdot \left(\frac{\partial c}{\partial b} \cdot \left(\frac{\partial b}{\partial a} \cdot \left(\frac{\partial a}{\partial x}\right)_1\right)_2\right)_3$$

1. $m = 1$; $F : \mathbb{R}^n \to \mathbb{R}$

$$\frac{\partial F}{\partial x} = \frac{\partial y}{\partial c} \cdot \frac{\partial c}{\partial b} \cdot \frac{\partial b}{\partial a} \cdot \frac{\partial a}{\partial x}$$

Dy_c

VYP

y^C_δ

y^δ_α

y^α_x

2. $m > n$, $F : \mathbb{R}^n \to \mathbb{R}^m$

m

y^y_c

d

y^φ_x

n

YVP

$\alpha : \mathbb{R}^n \to \mathbb{R}^k$